

From	To	Signal(s)		
host/top-level	key_commit_buffer	wr_en, wr_addr, wr_data, commit, zeroize		Notes:
key source (key_vault_rom or teammate block)	key_commit_buffer	key_hw[127:0]		The design has two planes: an unlock/control plane and an AES data plane.
key_commit_buffer	unlock_fsm	all_valid, match_q, commit_illegal		key_commit_buffer authenticates the attempted key and feeds unlock_fsm. unlock_fsm drives aes_enable.
host/top-level	unlock_fsm	commit, zeroize, reset, clk		AES operation should only be allowed when aes_enable && block_all_valid.
host/top-level	aes_input_block_buffer	blk_wr_en, blk_wr_addr, blk_wr_data, zeroize		AES output is captured by aes_result_buffer, and host reads it through rd_addr/rd_data/rd_valid.
aes_input_block_buffer	AES core	block_in[127:0]		
unlock_fsm + aes_input_block_buffer	AES start gating	aes_enable, block_all_valid		
AES core	aes_result_buffer	result_in[127:0], result_valid		
host/top-level	aes_result_buffer	rd_en, rd_addr, zeroize		
aes_result_buffer	host/top-level	rd_data, rd_valid, result_ready		

Rule	Expression	Meaning
Key check allowed	commit	user requests attempted key validation
Unlock success path	commit && all_valid && match_q across buffer/FSM	full key was provided and matched hardware key
Illegal key attempt	commit && !all_valid → commit_illegal	user tried to commit before writing all key words
AES operation allowed	aes_enable && block_all_valid	AES may only run when unlocked and full input block is present
Read result allowed	rd_en && result_ready	host may only read result when one has been captured
Trap condition	zeroize or illegal/mismatch path	system enters trapped state and AES is disabled

Signal	Width	Driven by	Consumed By	Meaning	Location
commit	1	host / top-level control	key_commit_buffer, unlock_fsm	user requests validation of the attempted key	
key_hw	128	key_vault_rom assembly logic or te	key_commit_buffer / comparator	stored hardware key used for authentication	
all_valid	1	key_commit_buffer	unlock_fsm	all 4 attempted key words have been written	
match_q	1	key_commit_buffer comparator pal	unlock_fsm	attempted key matches stored hardware key	
commit_illegal	1	key_commit_buffer	unlock_fsm	commit was asserted before the full 128-bit key had been written	
zeroize	1	host / external security control	all sensitive modules	clears sensitive state and forces trap behavior	
aes_enable	1	unlock_fsm	top-level AES start gating / AES c	permits AES operations only in unlocked state	
trap	1	unlock_fsm	top-level status / optional fault ha	system has entered sticky trapped state	
unlocked	1	unlock_fsm	top-level status	system is in unlocked state	
Note					
all_valid here is not the same as the all_valid in the AES data path					

Signal	Width	Driver	Consumer	Meaning
wr_en	1	host / top-level control	aes_input_block_buffer	write enable for AES input block buffer
wr_addr	2	host / top-level control	aes_input_block_buffer	selects which 32-bit input word is being written
wr_data	32	host / top-level control	aes_input_block_buffer	input word written into AES input buffer
block_in	128	aes_input_block_buffer	AES core / top-level AES data	assembled 128-bit AES input block
block_all_valid	1	aes_input_block_buffer	top-level AES control	all 4 AES input words have been written
aes_start	1	top-level AES control	AES core	starts AES operation when enabled and input block is complete
result_in	128	AES core	aes_result_buffer	128-bit AES output block
result_valid	1	AES core	aes_result_buffer	indicates AES output block is valid and should be captured
result_ready	1	aes_result_buffer	host / top-level read logic	a valid AES result has been captured and is available
rd_en	1	host / top-level control	aes_result_buffer	enables result word readout
rd_addr	2	host / top-level control	aes_result_buffer	selects which 32-bit result word to read
rd_data	32	aes_result_buffer	host / top-level read logic	selected 32-bit word of buffered AES result
rd_valid	1	aes_result_buffer	host / top-level read logic	indicates rd_data is valid in the current cycle
Note				
all_valid here is not the same as the all_valid in the unlock/control path				

Module	Purpose	Inputs	Outputs	Conencts to	Location
key_vault_rom	Stores protected hardware key	clk, rst_n, chip_en, read_en, zeroize, word_idx	data_out, oe, tamper_latched, read_once_latched	top-level key assembly logic or teammate keygen path	
key_activation_compare	Compares attempted key to hardware key	key_in[127:0], key_hw[127:0]	match	used inside key_commit_buffer	
key_commit_buffer	Collects 4x32-bit attempted key words and checks on commit	clk, reset, zeroize, wr_en, wr_addr, wr_data, commit, key_hw	key_in, valid_mask, all_valid, do_compare, match_q, commit_illegal	feeds unlock_fsm	
unlock_fsm	Controls locked/check/unlocked/trapped states	clk, reset, zeroize, commit, all_valid, match_q, commit_illegal	aes_enable, trap, unlocked	gates AES operation	
aes_input_block_buffer	Collects 4x32-bit AES input words into 128-bit block	clk, reset, zeroize, wr_en, wr_addr, wr_data	block_in, valid_mask, all_valid	feeds AES core input	
aes_core	Performs AES encryption/decryption	clk, reset, input block, start/control signals	result_out, result_valid	feeds aes_result_buffer	
aes_result_buffer	Captures AES result and supports 32-bit readback	clk, reset, zeroize, result_in, result_valid, rd_en, rd_addr	rd_data, rd_valid, result_ready	read by host/top-level	